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Pierfrancesco Martini

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Gender: Male **Date of birth:** 09/11/1999 **Nationality:** Italian

EDUCATION AND TRAINING

[01/11/2023 – Current]

PhD in Mathematics

Gran Sasso Science Institute https://www.gssi.it/education/mathematics-education

City: L'Aquila | **Country:** Italy | **Field(s) of study:** General Relativity, Analysis of PDEs, Mathematical Physics

[09/2021 – 27/10/2023]

Master Degree in Theoretical Physics, achieved with the completion of the faculty excellence path

Sapienza, Università di Roma https://www.uniroma1.it/it/pagina-strutturale/home

Address: Piazzale Aldo Moro 5, 00185, Rome, Italy | **Field(s) of study:** Natural sciences, mathematics and statistics: • *Physics* • *Mathematics* | **Final grade:** 11 0/110 cum Laude | **Thesis:** Topological transport in flat bands, advisor: Prof Gianluca Panati, co-advisor: Prof Domenico Monaco

[08/2018 – 21/10/2021]

Bachelor Degree in Physics, achieved with the completion of the faculty excellence path

Sapienza, Università di Roma https://www.uniroma1.it/it/pagina-strutturale/home

Address: Piazzale Aldo Moro 5, 00185, Rome, Italy | **Field(s) of study:** Natural sciences, mathematics and statistics: • *Physics* | **Final grade:** 110/110 cum Laude | **Thesis:** Study of the Migdal Effect via photoabsorption in light dark matter direct searches, advisor: Prof Andrea Messina, co-advisor: Dott Giovanni Grilli di Cortona

[08/2013 – 07/2018]

Scientific high school diploma

Liceo Scientifico M. V. Pollione, Avezzano (AQ)

Address: Via Aldo Moro 1, 67051, Avezzano, Italy | **Final grade:** 100/100 cum Laude

LANGUAGE SKILLS

Mother tongue(s): Italian

Other language(s):

English

LISTENING B2 **READING** C1 **WRITING** B2

SPOKEN PRODUCTION C1 **SPOKEN INTERACTION** B2

Levels: A1 and A2: Basic user; B1 and B2: Independent user; C1 and C2: Proficient user

SKILLS

Scientific programming in C | Scientific programming and basic data analysis in Python | Basic data analysis in R | LATEX for general purpose and scientific writing

ACADEMIC PURSUITS

[03/2023 – 07/2023] **Tutor of the course "Fundamentals of Mathematics part II (Linear Algebra and Vector Analysis)", for the degree course in Chemical Sciences, at Sapienza University**

Winner of the call n. 33/2022, Rep. n. 223/2022, Prot. n. 3138/V/1 for Master Degree students.

[09/2022 – 02/2023] **Tutor of the courses "Geometry" and "Mathematical Analysis", for the degree course in Civil and Industrial Engineering, at Sapienza University**

Winner of the call Rep. n. 476/2022, Prot. n. 2609/04-08-2022 for Master Degree students.

[03/2022 – 09/2022] **Tutor of the Course of "Physics I (Classical Mechanics with Laboratory)", for the degree course in Chemical Sciences, at Sapienza University**

Winner of the call DD N. 15/2022, Prot. n. 141/24-01-2022 for Master Degree students.

[03/2022 – 07/2022] **Laboratory Activity at INFN (Istituto Nazionale di Fisica Nucleare) regarding the Cygno project for Dark Matter Direct Search**

Collaboration with the team of the Cygno project: assembly, calibration and data analysis for the Lime-2 TPC prototype; development of a neural network for the position localization of signals from the TMPs response in the TPC.

Excellence path of the Master Degree in Physics

Extracurricular activities pursued in the excellence programs organized by the Faculty of Mathematical, Physical and Natural sciences:

- Geometry of Yang Mills Theories, with Professor Gianluca Panati
- K-Theories in Condensed Matter Physics, with Professor Domenico Monaco
- Topics in the Theory of Critical Phenomena, with Professor Sergio Caprara

Excellence path of the Bachelor Degree in Physics

Extracurricular activities pursued in the excellence programs organized by the Faculty of Mathematical, Physical and Natural sciences:

- Topics in Differential Geometry, with Professor Paolo Piccinni

- Dark Matter Direct Search, with Professor Gianluca Cavoto
- Green functions for the solution of PDEs and numerical implementation using Python, with Professor Daniele Barducci
- Advanced topics in Electromagnetism, with professor Francesco Lacava

FURTHER QUALIFICATIONS AND EXPERIENCES

[24/06/2026] **Blue Belt in Brazilian Jiu Jitsu under Sensei Mauro Zugaro**

[12/12/2016] **Black Belt (I Dan) in Goju Ryu Karate (FIJLKAM certification)**

TALKS, PRESENTATIONS AND SEMINARS

[18/03/2026] **Topics in General Relativity, second edition - Reading Seminar at GSSI**

Detailed exposition and comment of the paper "Nonlinear nature of gravitation and gravitational-wave experiments" by D. Christodoulou, intended for an audience already familiar with General Relativity.

[10/12/2025] **Theoretical Physics Journal Club - University of Cambridge, DAMTP**

PDEs and General Relativity: an introduction

The aim of this talk is to provide a gentle and hands-on introduction to the language of PDEs applied to General Relativity.

General Relativity provides to date our most accurate description of gravity. It combines a simple and concrete physical foundation with an incredibly rich mathematical structure. Since its formulation in 1915, huge progresses have been made concerning both the phenomenological and mathematical aspects of the theory. Unfortunately, these two research branches have parted ways over the years: as a result, there is a growing gap in their languages, in the problem they investigate and even in the reasons to investigate them.

This talk is motivated by the belief that bridging this gap, and hence having people who can handle both the foundational and the more concrete aspects of General relativity, is essential to achieve a deeper understanding of our Universe.

[05/12/2025] **Mathematical General Relativity Reading Group - University of Cambridge, DPMMS and DAMTP**

The Maxwell Equations on Schwarzschild--Tangherlini Black Holes: boundedness and decay of solutions

The aim of this talk is to show how to formulate the Maxwell equations for components of the Faraday--Maxwell tensor relative to the algebraically

special frame of a Myers--Perry spacetime, and how it is possible to use this form of the equations to discuss boundedness and stability of their solutions. We then specialise the system of equations to the Schwarzschild--Tangherlini case and use special decoupling properties properties of some of the electromagnetic components to prove uniform boundedness and decay of solutions.

[03/12/2025] **Junior Analysis Seminar - Imperial College London**

The Maxwell Equations on Myers–Perry Black Holes

Einstein's Equations of General Relativity can be formulated in more than (3+1) dimensions. Of particular interest both from the mathematical and astrophysical point of view is the conjectured nonlinear stability of (4+1)-dimensional stationary (spherical) vacuum black holes. Maxwell's Equations represent a natural intermediate model to be studied towards the comprehension of this problem: this talk will deal with some fundamental features, challenges and consequences of Maxwell's theory formulated on a (4 + 1)-dimensional Myers–Perry black hole spacetime.

Link: https://www.imperial.ac.uk/pure-analysis-and-pdes/seminars/jas/

[28/05/2025 – 29/05/2025] **Topics in General Relativity - Reading Seminar at GSSI, May 12th – May 30th, 2025**

Two 90 minutes lectures devoted to presenting and discussing the proof of Penrose's Incompleteness Theorem.

Link: https://sites.google.com/view/benomio/home/topics-in-general-relativity-2025?authuser=0



L'Aquila, 08/09/2026

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